

Coding & Solution

Date	5 July 2026
Team ID	SWTID-2026-4860
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Harsh-Saini-0819/EduGenie-Google-Gemini-Powered-Learning-Assistant.git
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	FastAPI, Jinja2 Templates, Hugging Face Transformers
Key Features Implemented	• Intelligent Q&A and Concept Simplification (Gemini 1.5 Pro) • Automated Quiz Generation & Learning Path Roadmap Builder • Fast, offline Text Summarization via local LaMini-Flan-T5 model
Pending / Incomplete Features	User session state saving, quiz database tracking, user authentication
Setup / Run Instructions	1. Activate venv & install requirements.txt 2. Set terminal variable \$env:GEMINI_API_KEY="your_key" 3. Launch local backend web server via uvicorn main:app --reload

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

EduGenie features a hybrid AI architecture optimized to deliver fast, low-overhead performance on standard consumer hardware. Built with an asynchronous FastAPI backend and server-side Jinja2 templates, it manages system latency without heavy frontend frameworks. The project strategically offloads complex analytical tasks—such as Q&A, quiz engineering, and personalized learning path - to the cloud-based Google Gemini 1.5 Pro API via the modern google-genai SDK. Meanwhile, it handles text summarization completely offline to preserve local privacy. This is achieved by utilizing an edge-based Hugging Face transformers pipeline loaded with the lightweight LaMini-Flan-T5-248M model.